

Learning compositional design rationale from synchronised audio and professional lighting control

"I just want to bring my intent into the machine."

The gap

Systems that turn music into light learn surface mappings: audio features to looks, colour values per frame, or control streams to control streams. They reproduce a style; they do not represent why a designer made a decision at a given moment in a song.

A professional lighting programme carries structure that spans the whole temporal form of the music. Recurrence, restraint, reveal, and the deliberate refusal to react to a beat are design decisions, and none of them live in the control values.

The corpus

101 5 1

show records designers production setup

Professional grandMA3 source-show records with native console semantics intact, synchronised to audio. Unlike control-stream or video-scraped corpora, cues, presets, recipes and phasers survive as authored objects.

All records share one luminaire topology. The collection constraint is reported as a methodological instrument: it alters behaviour and preserves semantic structure.

Decomposition

Source decomposition is complete across 101 records. It models maintained looks, animated layers, accent events, recurrence and ordered development. Native-output uncertainty and unresolved reactions remain explicit.

For RQ1, typed means heterogeneous object kinds: a trajectory, a semantic state, a relation and an event sequence are modelled as different objects.

Where this stands

Done. Source-supported decomposition and reproducibility checks; the academic layer contract drafted.

Prototype. Editable donor-transfer shows are under whole-song musical review.

Open. Published corpus audit and generator architecture: three candidates, no favourite; resolved by a preregistered pilot plus method contract.

Next. Representation round trip, held-out-song generation and the matched hierarchy diagnostic.

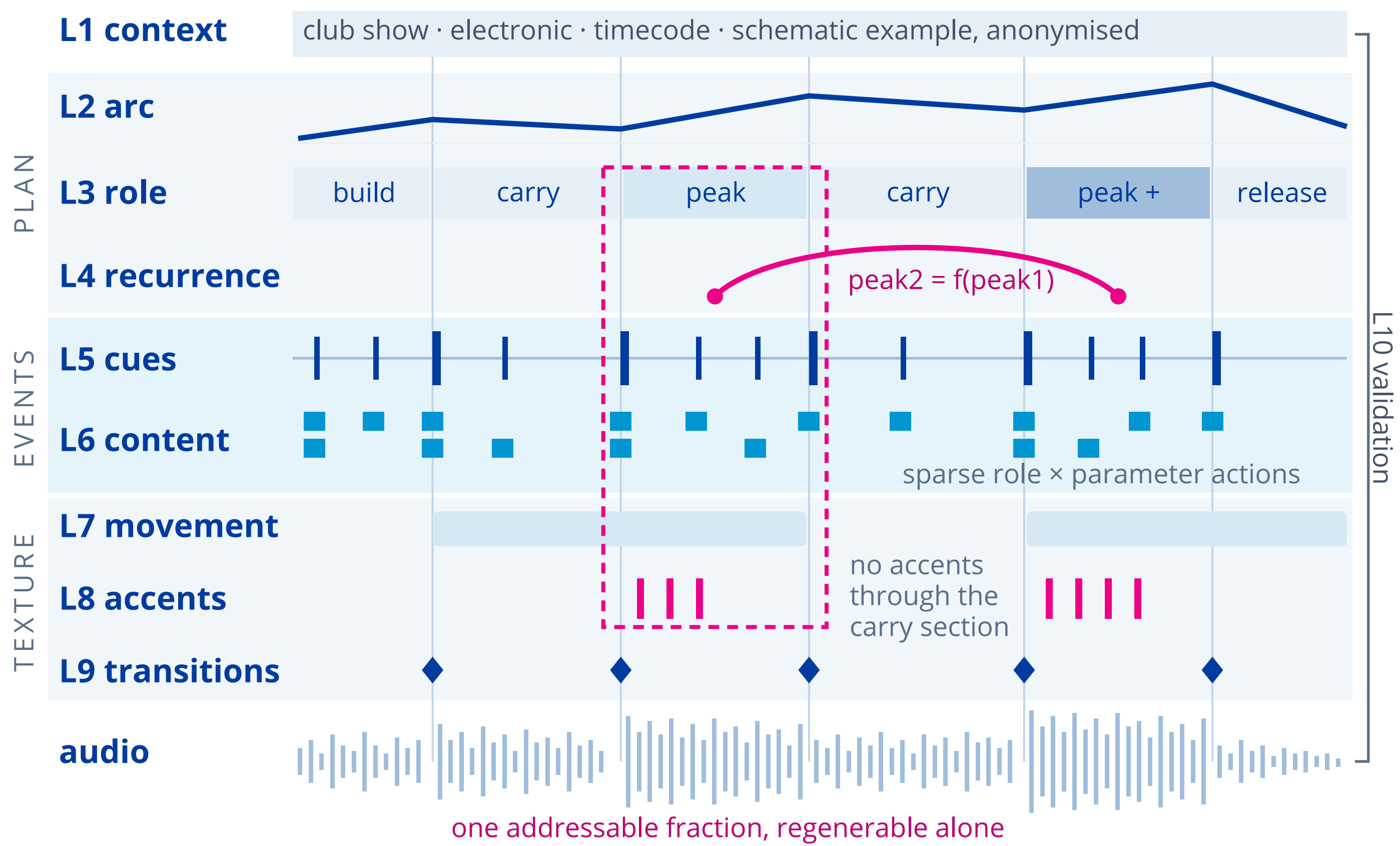


Fig. 1: Proposed layered representation above synchronised audio, shown schematically. Nine layers in three bands (plan, events, texture), each with its own object kind. The dashed region marks a selected temporal fraction; regeneration should preserve protected content. Addressability is a requirement to be tested.

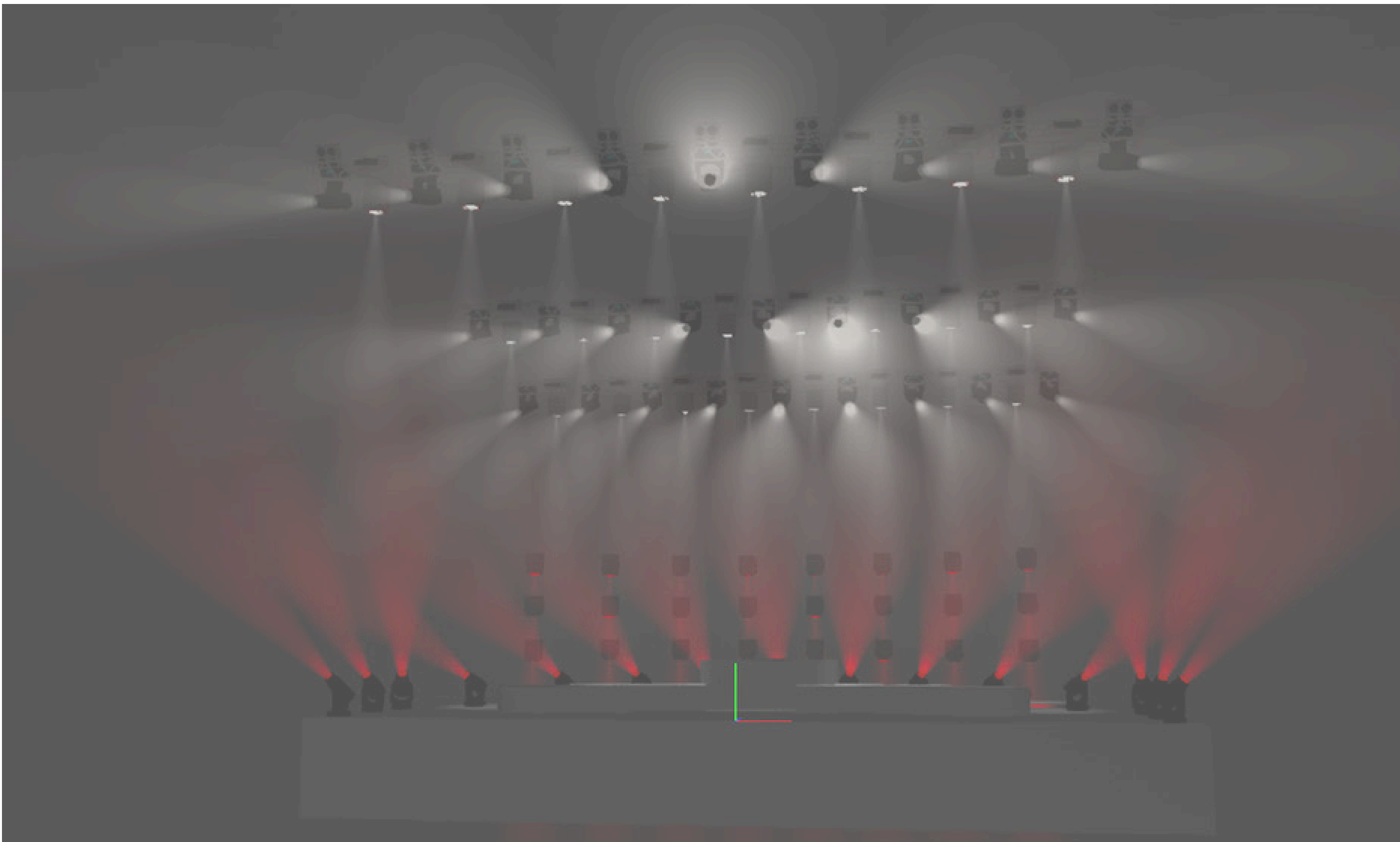


Fig. 2: Illustrative image; prototype capture to follow. An archived visualizer image temporarily occupies this replaceable image slot.

The research question

Can a hierarchy-aware audio-conditioned model generate an addressable, typed representation of professional concert-lighting structure and console semantics for held-out songs, within a fixed production environment, such that selected layers or temporal fractions can be regenerated while protected content stays stable?

Representational adequacy. Authored shows encode, reconstruct from the typed representation under a declared bottleneck.

Hierarchy utility. Typed dependencies beat capacity-matched flat and collapsed alternatives.

Preliminary source evidence.

101 records yield 1,338 sections, 3,164 look-bearing main cues, 12,765 accent events and 569 within-show recurrence families. These counts motivate modelling multiple scales; they are not independent training examples. Academic audit pending.

Predictive adequacy. The representation is generated from held-out audio, structure conditioning, and context.

Intervention integrity. Selected layers or fractions regenerate while declared protected content stays stable.

Proposed generation

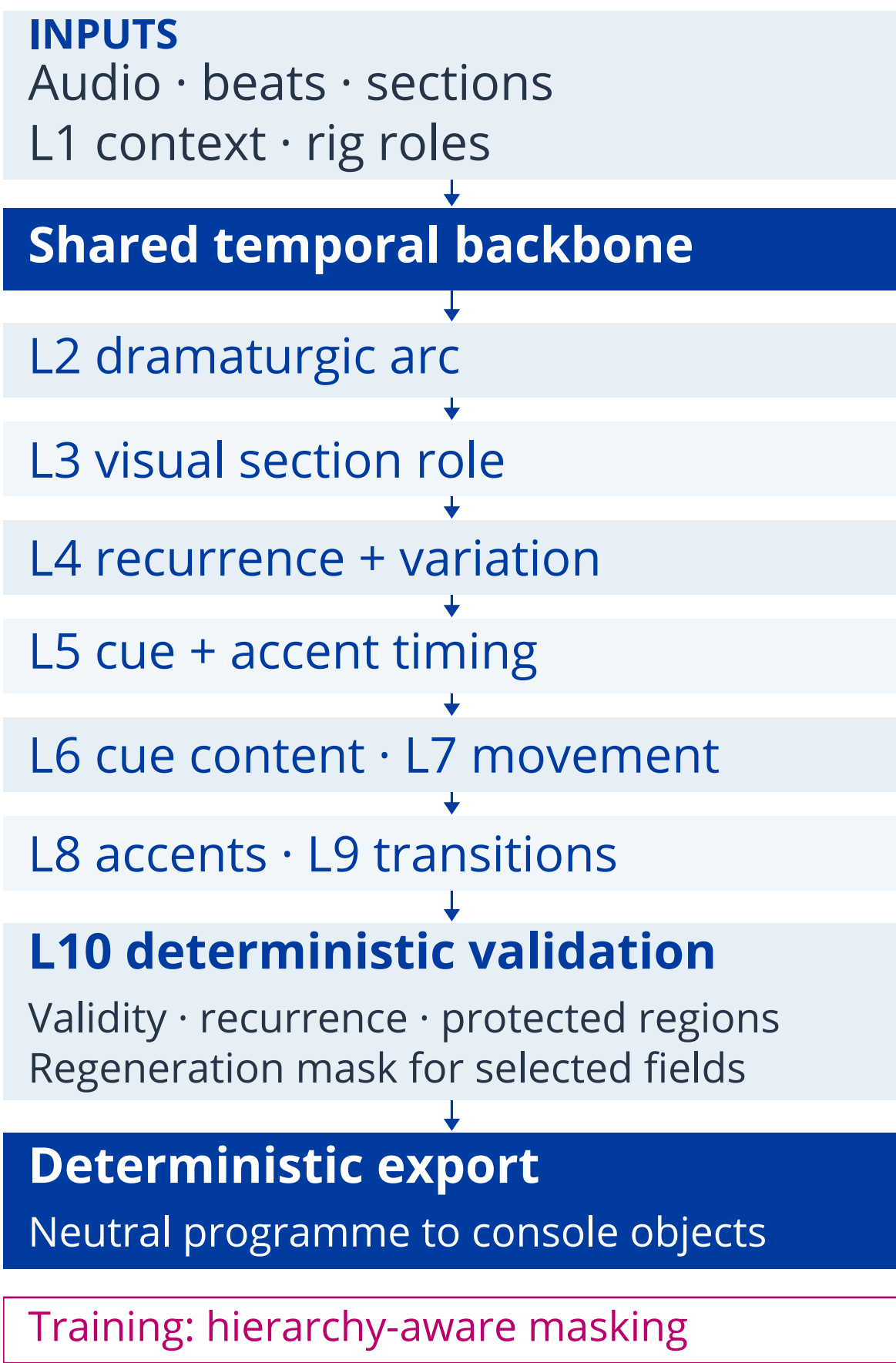


Fig. 3: Proposed generation architecture. Typed heads share a masked objective for generation and editing. Validation and export are deterministic.

Planned diagnostic [fixed]

Three variants, matched on capacity, input, supervision and output, so the only difference is the dependency structure.

Flat. One encoder predicts all fields in parallel, with losses on every intermediate target: it lacks the typed conditional dependency, not the supervision.

Collapsed. Section-level plan first, all realisation second.

Fully typed. The explicit dependency graph.

How it will be judged

Song-grouped splits throughout: no normalisation, threshold or prior touches held-out material. A named vector of measurements rather than one scalar. Thresholds fixed before results are seen, and a permanent hold-out excluded from any optimisation.

Several professional outputs can be valid for the same audio: reference similarity, distributional plausibility, structural alignment and validity are reported separately.

Not claimed

Designers' actual reasons. Artistic merit. Practitioner acceptance. Human-valued editability. Generalisation to a broad designer population. Machine-learning compositional generalisation. Learned cross-rig generalisation.

Poster, project & contact

github.com/twotonetobi/rq1-lighting-representation

Preliminary poster
22 September 2026

